Project - High Level Design

on

Healthcare Planning Assistant Agent

Course Name: DevOps

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: DO-22

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Academic Year: 2025-26

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1. Introduction

1.1 Scope of the Document

This document provides a High-Level Design (HLD) for the Healthcare Planning Assistant system. It describes the system architecture, components, workflows, data handling, and integration with external services. The document serves as a blueprint for developers, testers, and stakeholders to understand the system design and its functionalities.

1.2 Intended Audience

- Developers and Software Engineers
- System Architects
- QA/Test Engineers
- DevOps Engineers
- Project Evaluators (Academic/Industry)

1.3 System Overview

The Healthcare Planning Assistant is a FastAPI-based web application that allows users to input symptoms and receive AI-generated healthcare guidance. It uses LLMs (Google Gemini / Groq LLaMA) via LangChain to generate structured medical advice in four sections:

- Possible Conditions
- Precautions
- Medication Guidance
- Nutrition Advice

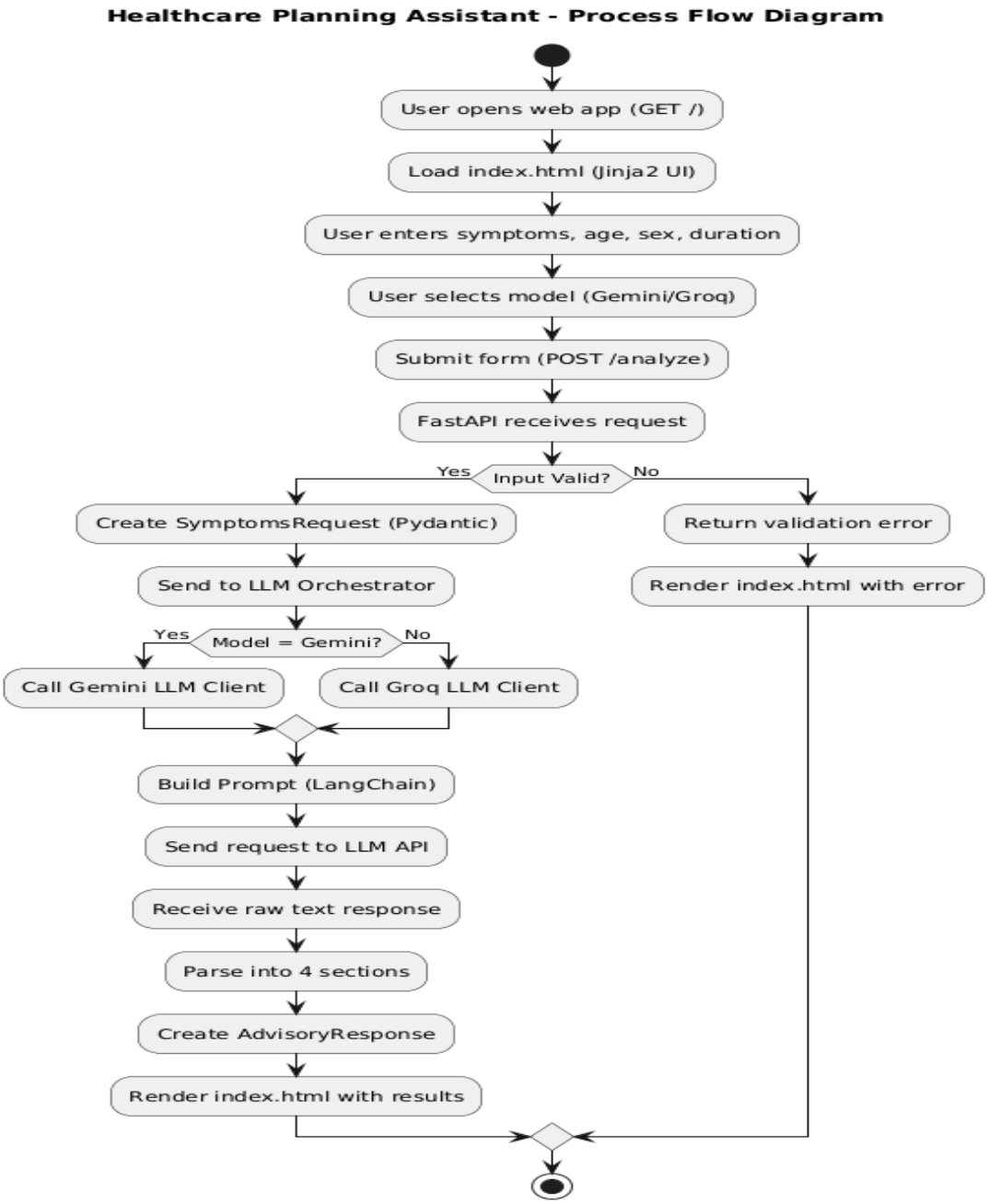
2. System Design

2.1 Application Design

The application follows a **layered architecture**:

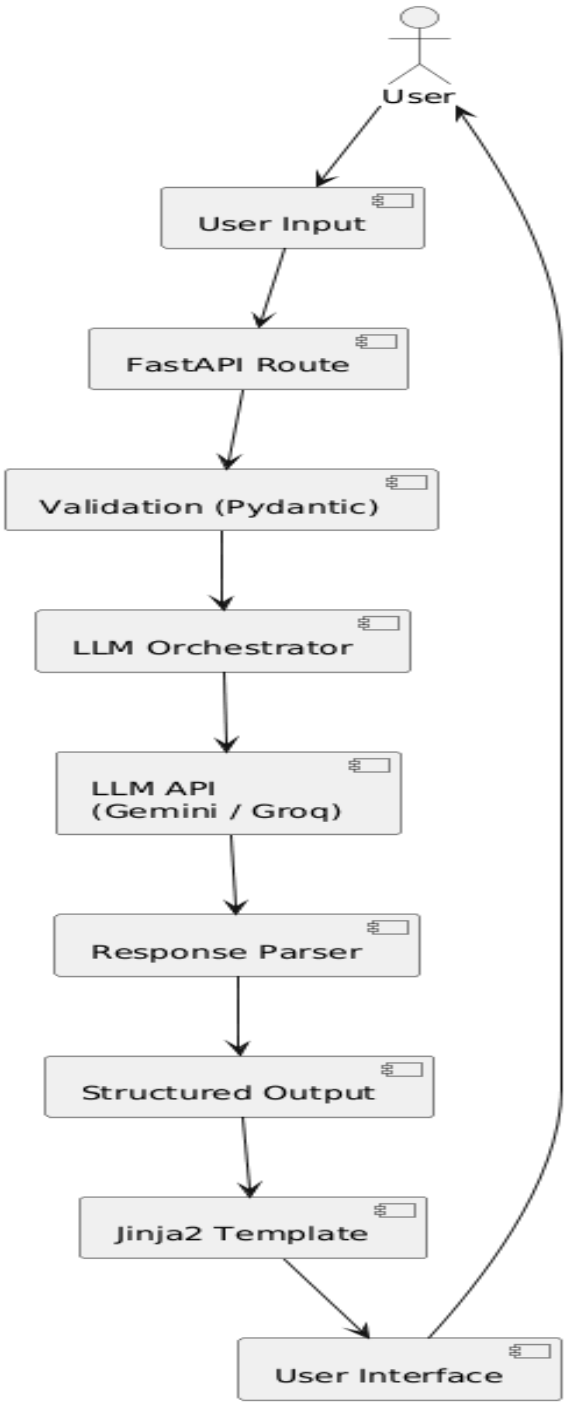
- **Presentation Layer**: Jinja2 templates + Bootstrap UI
- **API Layer**: FastAPI routes
- **Service Layer**: LLM orchestration
- **Domain Layer**: Pydantic models (DTOs)
- **Configuration Layer**: Environment-based settings

2.2 Process Flow



2.3 Information Flow

2.3 Information Flow - Healthcare Planning Assistant



2.4 Components Design

Frontend (UI)

- Jinja2 templates (HTML rendering)
- Bootstrap for styling

Backend

- FastAPI framework
- Routing module (`routes_symptoms.py`)

Service Layer

- LLM Orchestrator
- Gemini Client
- Groq Client

Configuration

- Environment variables (.env)
- Settings loader

External Systems

- Google Gemini API
- Groq LLaMA API

2.5 Key Design Considerations

- **Stateless Architecture** (No database)
- **Modular Design** (Separation of concerns)
- **Provider Flexibility** (Gemini / Groq switch)
- **Error Handling & Validation**
- **Security of API Keys (user-provided at runtime)**
- **Scalability via containerization**

2.6 API Catalogue

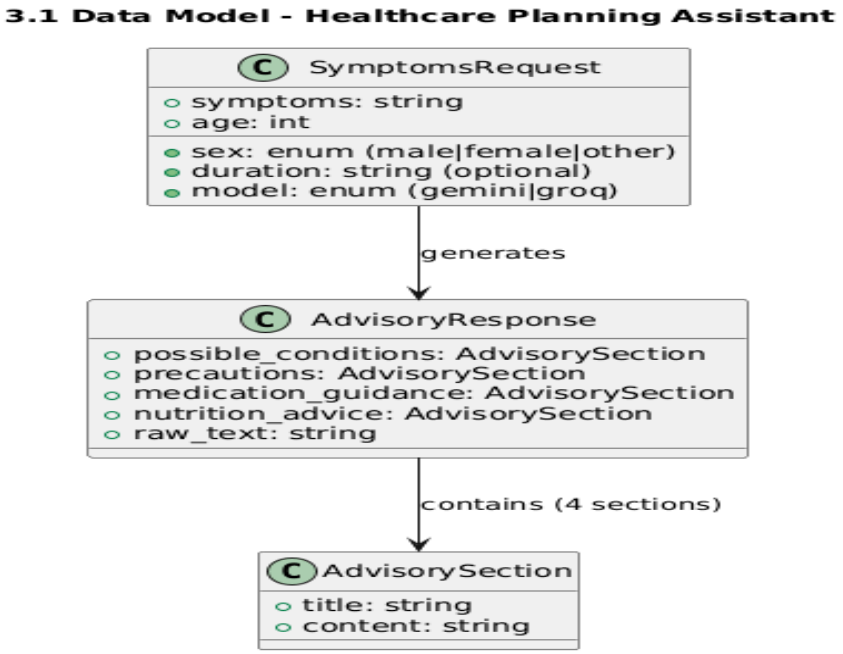
Endpoint	Method	Description
`/`	GET	Load homepage
`/analyze`	POST	Analyze symptoms
`/health`	GET	Health check

3. Data Design

3.1 Data Model

No persistent database is used. Data is handled via in-memory DTOs:

- **SymptomsRequest**
- **AdvisoryResponse**



3.2 Data Access Mechanism

- Data is passed via HTTP requests (form submission)
- Processed in-memory using Pydantic models
- No database queries involved

3.3 Data Retention Policies

- No data is stored permanently
- API keys are entered at runtime and not saved
- Responses exist only during request lifecycle

3.4 Data Migration

- Not applicable (no persistent storage)

4. Interfaces

- **User Interface**: Web-based HTML form
- **API Interface**: REST endpoints (FastAPI)
- **External APIs**:
  - Gemini API
  - Groq API

5. State and Session Management

- The system is **stateless**
- No session storage or cookies used
- Each request is independent

6. Caching

- Currently not implemented
- Future enhancement:
  - Cache repeated queries
  - Use Redis for performance optimization

7. Functional Requirements

7.1 User Authentication & Authorization

- Not implemented in current version
- Future scope: login/signup system

7.2 Infrastructure Configuration Management

- Managed via `.env` file
- Includes API keys and model settings

7.3 Provisioning Control & Monitoring

- Health endpoint `/health`
- Logs for debugging

7.4 Telecom Service Orchestration

- Not applicable to this system

7.5 Dockerfile (Application Containerization)

- Application can be containerized using Docker
- Ensures portability and consistency

7.6 GitHub Actions (Workflow Configuration)

- Automates build, test, and deployment
- Ensures CI pipeline

7.7 CI/CD Strategy

- Continuous Integration: Code validation & testing
- Continuous Deployment: Auto deployment to server/cloud

8. Non-Functional Requirements

8.1 Security Aspects

- API keys are not stored or logged
- Input validation using Pydantic
- CORS configuration
- Protection against invalid inputs

8.2 Performance Aspects

- FastAPI ensures high performance
- Async LLM calls improve efficiency
- Lightweight architecture (no DB overhead)

9. References

- FastAPI Documentation
- LangChain Documentation
- Google Gemini API Docs
- Groq API Docs
- Bootstrap Documentation
- Group Name: **11D6- Healthcare Planning Assistant Agent**

10. Github

- Github link - https://github.com/MeetSolanki0/Dockerized-Healthcare-Service